

**SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class
Test**

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

**COURSE OUTCOME
COVERED**

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I S.Varsha Reddy(192572177) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:S.Varsha Reddy

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Answer:

When an image is too bright, the most suitable grey level transformation is the Negative Transformation (or, if specifically reducing high intensity while preserving detail, Power-Law/Gamma Correction with $\gamma > 1$; in most introductory Digital Image Processing questions, Negative Transformation is commonly expected only for special cases, while Gamma Correction is the correct choice for brightness adjustment).

Concept:

Power-Law (Gamma) Transformation modifies the intensity values of an image using the formula:

$$s = c \times r^\gamma$$

where:

- r = input pixel intensity,
- s = output pixel intensity,
- c = constant,
- γ (gamma) = gamma value.

For an image that is too bright, choose $\gamma > 1$. This decreases high-intensity

pixel values, making the image darker and improving visibility.

Application Process:

1. Read the input image.
2. Select a gamma value greater than 1.
3. Apply the Power-Law transformation to every pixel.
4. Generate the enhanced output image.

Expected Outcome and Justification:

The transformation reduces excessive brightness while preserving important image details. It improves contrast in bright regions, making objects and edges easier to identify. Therefore, Gamma Correction ($\gamma > 1$) is the most suitable grey level transformation for correcting an overly bright image.

Result:

The output image has balanced brightness, improved contrast, and better visual quality, making it easier for analysis and interpretation.

2.A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Answer:

A suitable grey level transformation for enhancing a dark image is the Logarithmic (Log) Transformation.

The Log Transformation is represented by the formula:

$$s = c \log (1+r)$$

where:

- r = input pixel intensity,

- s = output pixel intensity,
- c = scaling constant.

Explanation

Dark images contain most of their pixel values in the low-intensity (dark) range, making important details difficult to observe. The Log Transformation expands these low-intensity values while compressing high-intensity values. As a result, small differences between dark pixels become more visible.

Justification

- Enhances hidden details in shadow or dark regions.
- Improves the overall brightness without excessively increasing already bright areas.
- Produces better contrast in low-intensity regions, making objects easier to identify.
- Preserves image information while improving visual quality.

Example

Consider a nighttime road image where pedestrians and vehicles are difficult to see. After applying Log Transformation, the darker regions become brighter, revealing hidden objects and improving visibility while preventing bright areas such as streetlights from becoming excessively bright.

Conclusion

Therefore, Logarithmic Grey Level Transformation is the most appropriate technique for enhancing a dark image because it effectively increases the visibility of low-intensity regions, improves contrast, and reveals hidden details, making the image clearer for visual interpretation and further image processing.

3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and

illustrate how it improves image quality with logical reasoning.

Answer:

A suitable histogram-based technique for improving the contrast of an image is Histogram Equalization.

Explanation

Histogram Equalization is an image enhancement technique that redistributes the pixel intensity values so they are spread more evenly across the entire grey-level range (0–255). This increases the difference between dark and bright regions, making hidden details easier to observe.

How It Works

1. Compute the histogram of the input image.
2. Calculate the cumulative distribution function (CDF).
3. Map the original pixel values to new intensity values using the CDF.
4. Generate an enhanced image with improved contrast.

Justification

- In a low-contrast image, most pixel values are concentrated within a narrow intensity range.
- Histogram Equalization spreads these pixel values over the full intensity range.
- This improves the visibility of edges, textures, and fine details.
- It enhances the overall appearance of the image without changing its important information.

Example

Consider a foggy outdoor image where buildings and trees appear dull and unclear. After applying Histogram Equalization, the intensity values are redistributed, making the buildings, trees, and road edges sharper and easier .

Conclusion

Histogram Equalization is an effective histogram-based contrast enhancement technique because it redistributes pixel intensities over the full grey-level range, improves contrast, reveals hidden details, and produces a clearer and more visually informative image. This makes it widely used in medical imaging, satellite images, and digital photography.

4.A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Answer:

The most suitable method for achieving a uniform intensity distribution in a low-contrast image is Histogram Equalization.

Concept

Histogram Equalization is a histogram-based image enhancement technique that redistributes the grey-level intensity values so that they are spread more uniformly across the available intensity range (0–255). This improves the contrast and reveals details that are difficult to see in the original image.

Working Process

1. Calculate the histogram of the input image.
2. Compute the cumulative distribution function (CDF) from the histogram.
3. Use the CDF to map the original pixel intensities to new intensity values.
4. Generate the enhanced image with a more uniform distribution of grey levels.

Justification

- Low-contrast images have pixel values concentrated within a small intensity range.
- Histogram Equalization spreads these values across the full intensity range.
- This increases the difference between neighbouring intensity levels, making edges, textures, and fine details more visible.
- The resulting histogram becomes more evenly distributed, leading to better image quality.

Example

A medical X-ray image may appear dull because most pixels have similar intensity values. After applying Histogram Equalization, bones, tissues, and other structures become more distinct, helping doctors identify important features more accurately.

Conclusion

Histogram Equalization is the best method for enhancing a low-contrast image because it redistributes pixel intensities uniformly, improves contrast, enhances hidden details, and produces a clearer, more informative image for analysis and interpretation.

5.An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Answer:

The most appropriate enhancement technique for correcting non-uniform illumination is Adaptive Histogram Equalization (AHE) or its improved version, Contrast Limited Adaptive Histogram Equalization (CLAHE).

Explanation

Unlike standard Histogram Equalization, which enhances the entire image uniformly, CLAHE divides the image into small regions (tiles) and enhances the contrast of each region separately. This corrects uneven brightness and improves local details without over-enhancing noise.

Working Process

1. Divide the image into small non-overlapping tiles.
2. Compute the histogram for each tile.
3. Apply histogram equalization to each tile individually.
4. Limit the contrast to avoid excessive amplification of noise (Contrast Limiting).
5. Merge the enhanced tiles smoothly using interpolation to produce the final image.

Justification

- Corrects uneven illumination in different parts of the image.
- Enhances local contrast and reveals hidden details.
- Prevents over-brightening and reduces noise amplification.
- Produces a natural-looking image with balanced brightness.

Example

Consider an X-ray image where one side appears darker than the other. Applying CLAHE enhances each region independently, making bones and tissues clearly visible across the entire image while maintaining a uniform brightness level.

Conclusion

Contrast Limited Adaptive Histogram Equalization (CLAHE) is the most suitable technique for correcting non-uniform illumination. It enhances local contrast, balances brightness across different regions, preserves image details, and minimizes noise, resulting in a clearer and more informative

image for analysis.

6.A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Answer:

A Median Filter is the most suitable smoothing filter for removing random intensity variations (especially salt-and-pepper noise) while preserving image edges.

Definition

A Median Filter is a non-linear spatial smoothing filter that replaces the value of each pixel with the median value of its neighboring pixels instead of the average.

Working Mechanism

1. A small window (e.g., 3×3 or 5×5) is moved over the image.
2. The pixel values inside the window are arranged in ascending order.
3. The median (middle) value is selected.
4. The center pixel is replaced with this median value.
5. The process is repeated for every pixel in the image.

Example:

Window values:

12, 14, 15, 18, 20, 150, 21, 19, 17

Sorted values:

12, 14, 15, 17, 18, 19, 20, 21, 150

Median = 18

The noisy pixel 150 is replaced with 18, effectively removing the noise.

Impact on Noise Reduction

- Removes salt-and-pepper (impulse) noise effectively.
- Preserves edges and fine details better than averaging filters.
- Reduces random intensity variations without causing significant image blur.
- Improves the visual quality of the image, making it suitable for further image processing tasks such as segmentation and feature extraction.

Justification

The Median Filter is preferred because it removes random noise while maintaining important image boundaries. Unlike the Mean Filter, it does not blur edges significantly, making it the best choice for images affected by impulse noise.

Conclusion

Thus, the Median Filter is the most suitable smoothing filter for noisy images with random intensity variations because it provides effective noise reduction, preserves edges, and enhances overall image quality, making it ideal for digital image processing applications.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Answer:

A Bilateral Filter is the most appropriate filtering approach for reducing noise while preserving important image details and edges.

Definition

A Bilateral Filter is a non-linear edge-preserving smoothing filter that considers both the spatial distance between pixels and the difference in pixel intensity. This allows it to smooth similar pixels while preventing blurring across edges.

Working Mechanism

1. A neighborhood window (e.g., 5×5) is selected around each pixel.
2. Each neighboring pixel is assigned a weight based on:
 - o Spatial closeness (nearby pixels have higher weight).
 - o Intensity similarity (pixels with similar intensity have higher weight).
3. The center pixel is replaced by the weighted average of these neighboring pixels.
4. Pixels across edges receive very low weights, so edges are preserved while noise is reduced.

Impact on Noise Reduction

- Effectively removes Gaussian and random noise.
- Preserves edges, textures, and fine details.
- Produces smoother images without excessive blurring.
- Improves image quality for further tasks such as segmentation, feature extraction, and object recognition.

Justification

The Bilateral Filter is preferred because it smooths homogeneous regions while preserving sharp edges. Unlike the Mean Filter or Gaussian Filter, it does not blur object boundaries, achieving an excellent balance between noise removal and detail preservation.

Conclusion

Thus, the Bilateral Filter is the most suitable filtering approach for images that require effective noise reduction without significant loss of details. It provides high-quality smoothing while maintaining important image structures, making it widely used in digital image processing.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Answer:

An Unsharp Masking (High-Boost Filtering) technique is the most suitable sharpening method for restoring details in a blurred image after smoothing.

Definition

Unsharp Masking is a sharpening technique that enhances image edges by adding the high-frequency details (edges) back to the original image. It improves image sharpness without significantly affecting the overall brightness.

Working Principle

1. The original image is first blurred using a smoothing filter (e.g., Gaussian filter).
2. The blurred image is subtracted from the original image to obtain an edge mask containing high-frequency details.
3. The edge mask is added back to the original image.
4. The resulting image has sharper edges and clearer details.

Impact on Image Quality

- Restores lost edge details after smoothing.

- Enhances image sharpness and clarity.
- Improves the visibility of fine textures and object boundaries.
- Maintains overall brightness while making important features more distinct.

Justification

Unsharp Masking is preferred because it selectively enhances edges rather than the entire image. Unlike simple high-pass filtering, it provides controlled sharpening with fewer unwanted artifacts, making it ideal for restoring details in images blurred by smoothing operations.

Conclusion

Thus, Unsharp Masking (High-Boost Filtering) is the most suitable sharpening method for restoring details in blurred images. It effectively enhances edges, improves visual quality, and preserves important image information, making it widely used in digital image processing.

9. **Fine details in an image are not clearly visible.**

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Answer:

A High-Pass Filter (HPF) is the most suitable filtering technique for enhancing fine details in an image because it emphasizes the high-frequency components, which represent edges, textures, and fine image details.

Definition

A High-Pass Filter is a sharpening filter that allows high-frequency components to pass while suppressing low-frequency components such as smooth regions and gradual intensity variations.

Working Principle

1. A high-pass filter mask (such as the Laplacian or High-Pass Kernel) is applied to the image.
2. The filter detects rapid intensity changes between neighboring pixels.
3. Low-frequency (smooth) regions are reduced, while high-frequency information (edges and fine details) is enhanced.
4. The processed image appears sharper and more detailed.

Impact on High-Frequency Components

- Enhances edges, textures, and fine details.
- Improves the visibility of small image features.
- Increases image sharpness and contrast around object boundaries.
- May slightly amplify noise if the image contains excessive high-frequency noise.

Justification

The High-Pass Filter is preferred because it highlights important image details by strengthening high-frequency components while reducing smooth background information. This makes it highly effective for image enhancement and sharpening applications.

Conclusion

Thus, the High-Pass Filter is the most suitable technique for enhancing fine details in an image. By emphasizing high-frequency components, it restores sharpness, improves edge visibility, and enhances the overall quality of the image, making it useful in digital image processing.

10.A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Answer:

A Median Filter is the most suitable spatial filtering method for preprocessing a noisy image before further analysis because it effectively removes noise while preserving image edges and important details.

Definition

A Median Filter is a non-linear spatial filter that replaces the value of each pixel with the median value of its neighboring pixels. It is especially effective for removing salt-and-pepper (impulse) noise.

Working Mechanism

1. Select a neighborhood window (e.g., 3×3 or 5×5) around each pixel.
2. Arrange all pixel values within the window in ascending order.
3. Find the median (middle) value.
4. Replace the center pixel with the median value.
5. Repeat the process for every pixel in the image.

Example:

Window values:

15, 18, 20, 22, 25, 180, 24, 21, 19

Sorted values:

15, 18, 19, 20, 21, 22, 24, 25, 180

Median = 21

The noisy pixel 180 is replaced with 21, effectively removing the noise.

Impact on Image Preprocessing

- Removes salt-and-pepper (impulse) noise efficiently.
- Preserves edges and fine details.
- Improves image quality without excessive blurring.
- Produces a cleaner image for segmentation, feature extraction, and object recognition.

Justification

The Median Filter is the preferred spatial filtering method because it removes noise while maintaining image boundaries. Unlike the Mean Filter, it does not significantly blur edges, making it ideal as a preprocessing step before further image analysis.

Conclusion

Thus, the Median Filter is the most suitable spatial filtering technique for preprocessing noisy images. It effectively reduces noise, preserves important details, and improves the accuracy of subsequent image processing operations.